HISTOPATHOLOGICAL SPECTRUM OF UPPER GI: INSIGHTS FROM SAMPLED CASES IN KENYA.

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ANATOMIC PATHOLOGIST
KUTRRH

CASE 1

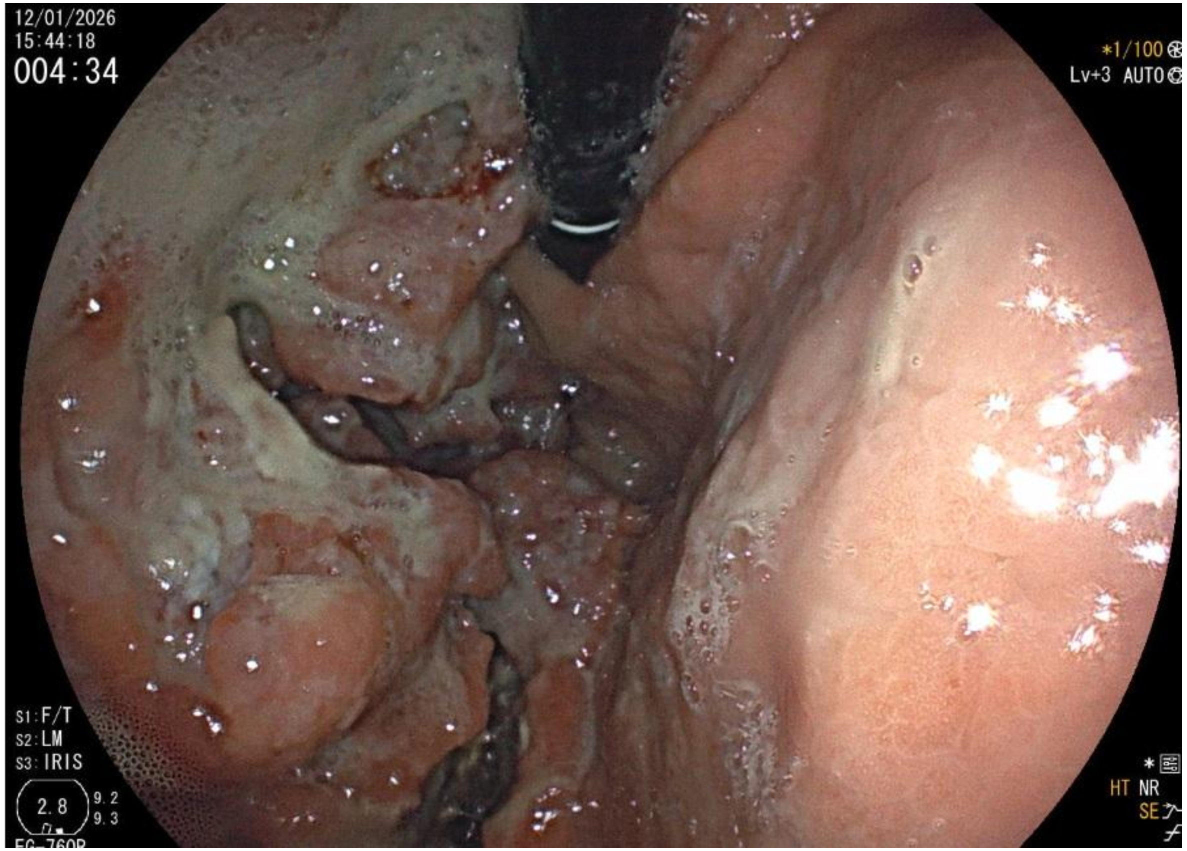
SM

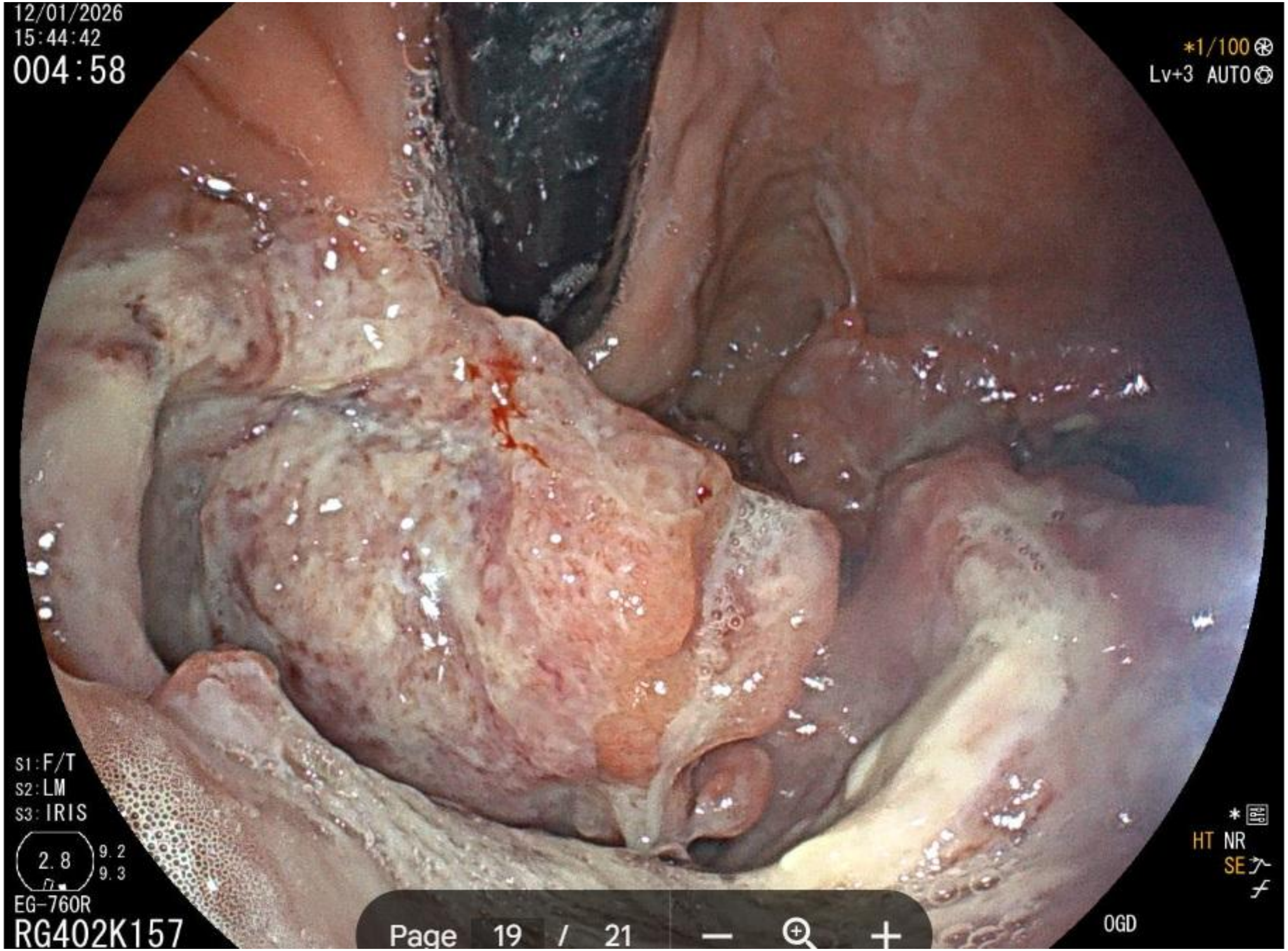
41yr old Male

Presented with abdominal distension

Endoscopic findings of exophytic gastric mass with ulceration.

ENDOSCOPIC FINDING

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EG-760R
RG402K157
HT NR
SE
OGD
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S1:F/T
S2:LM
S3:IRIS
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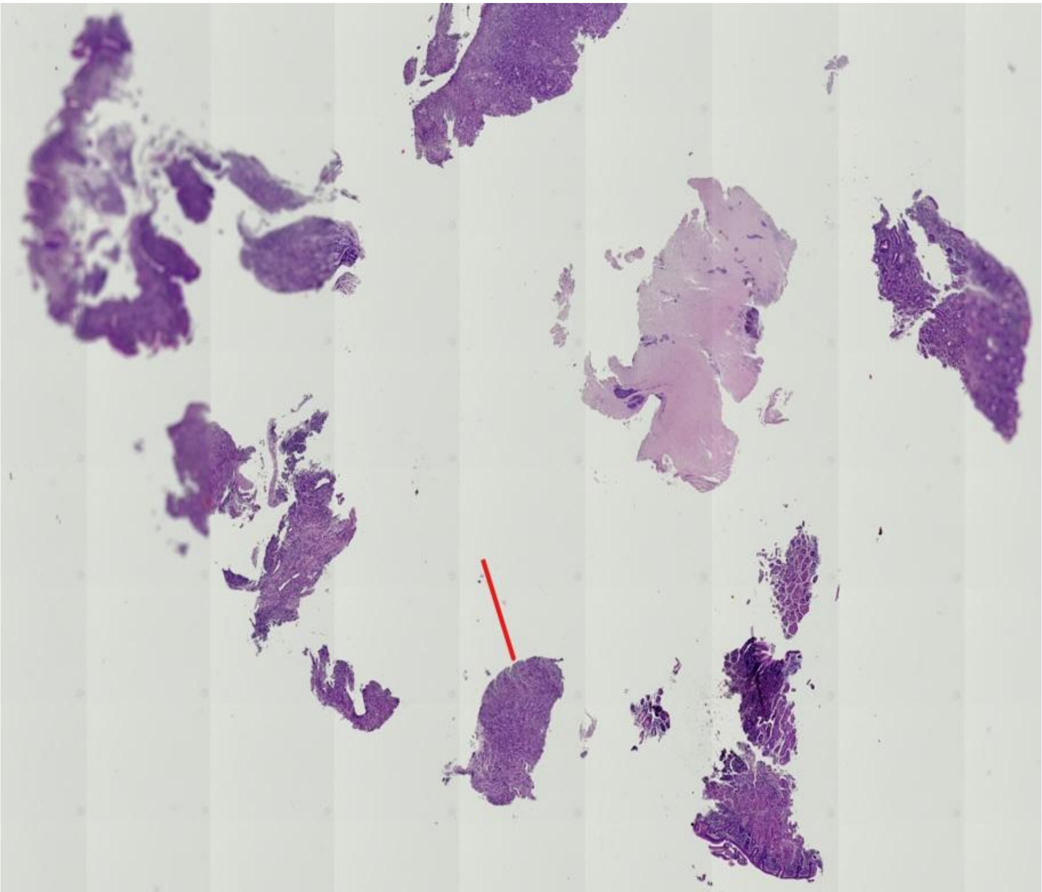
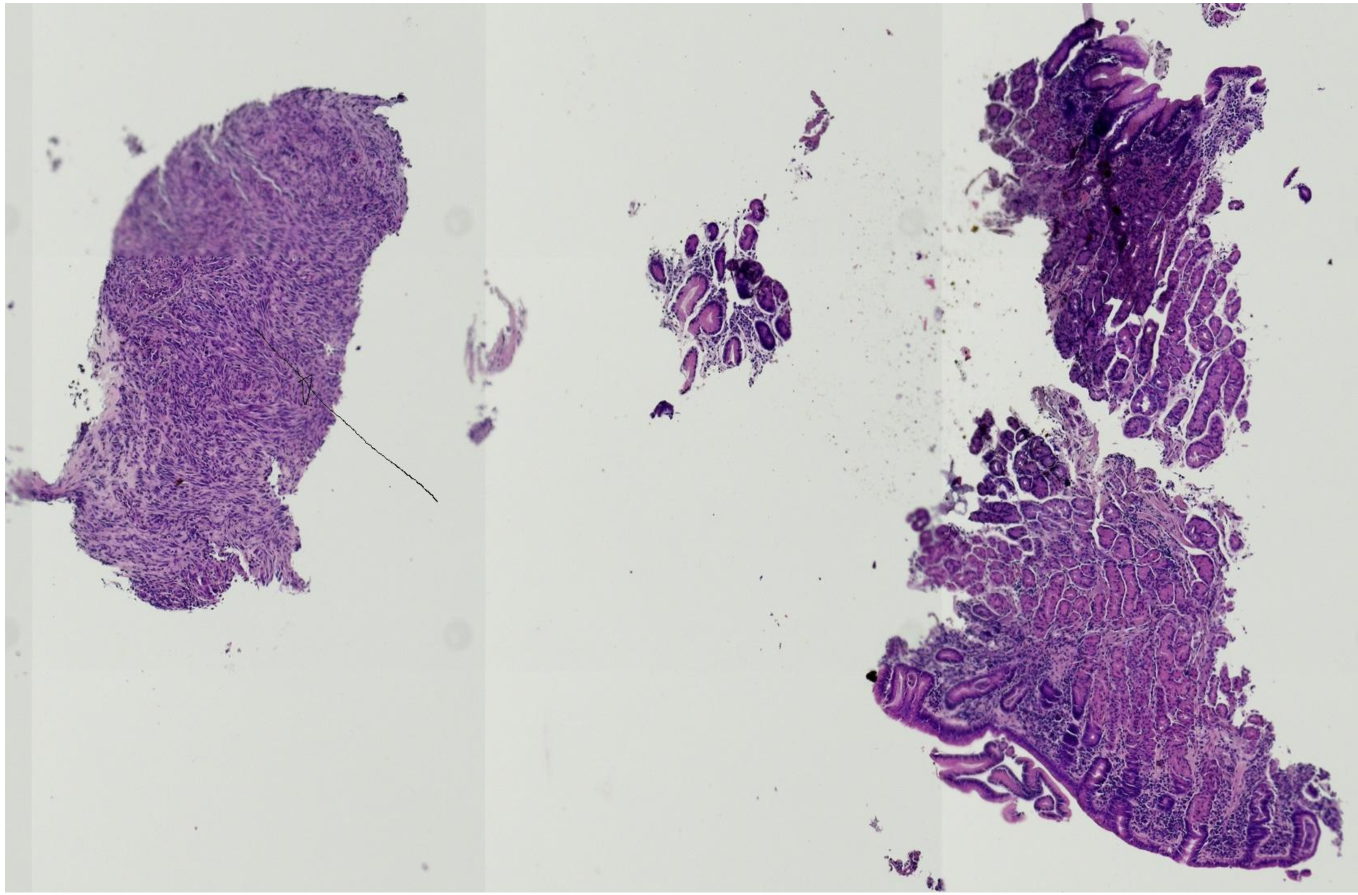
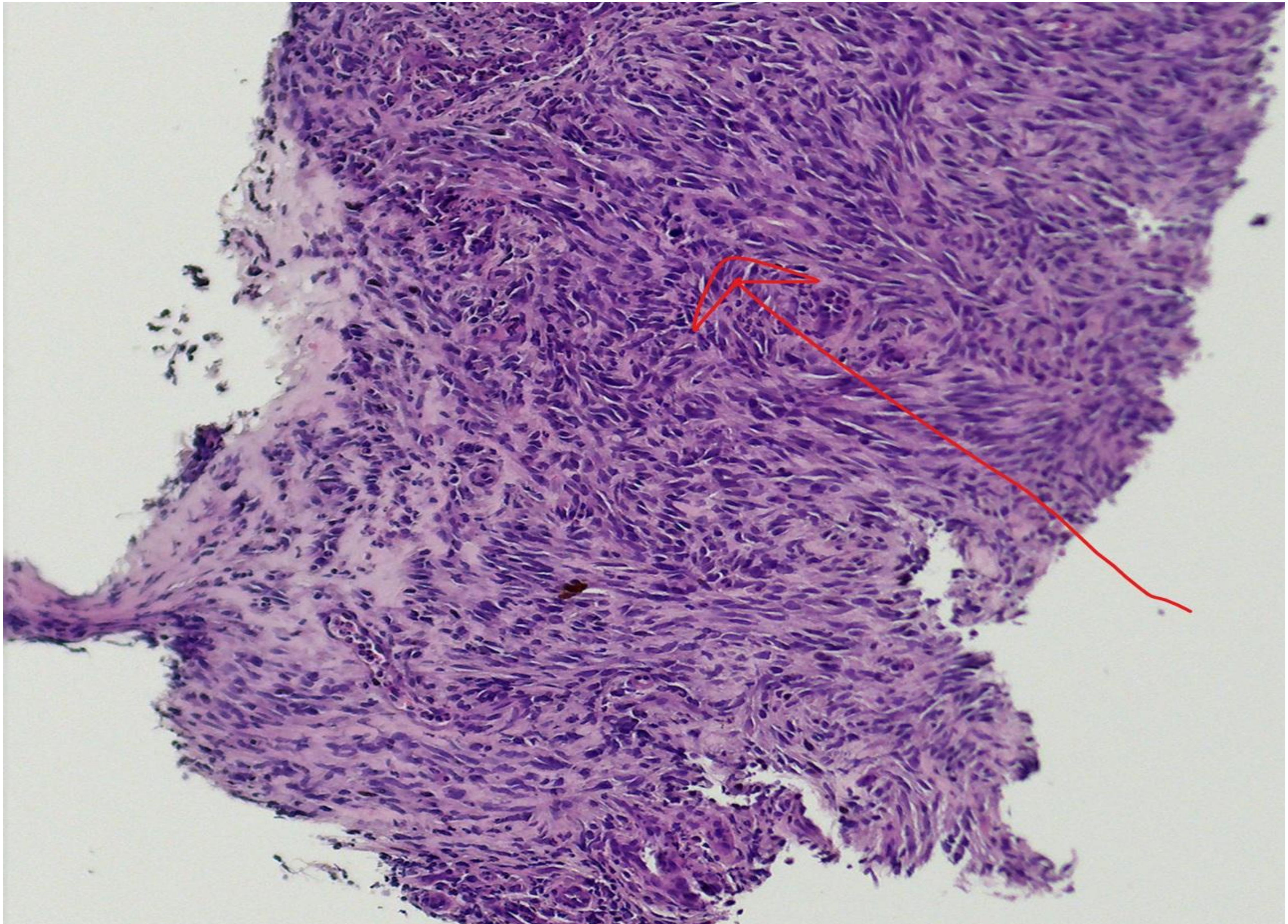
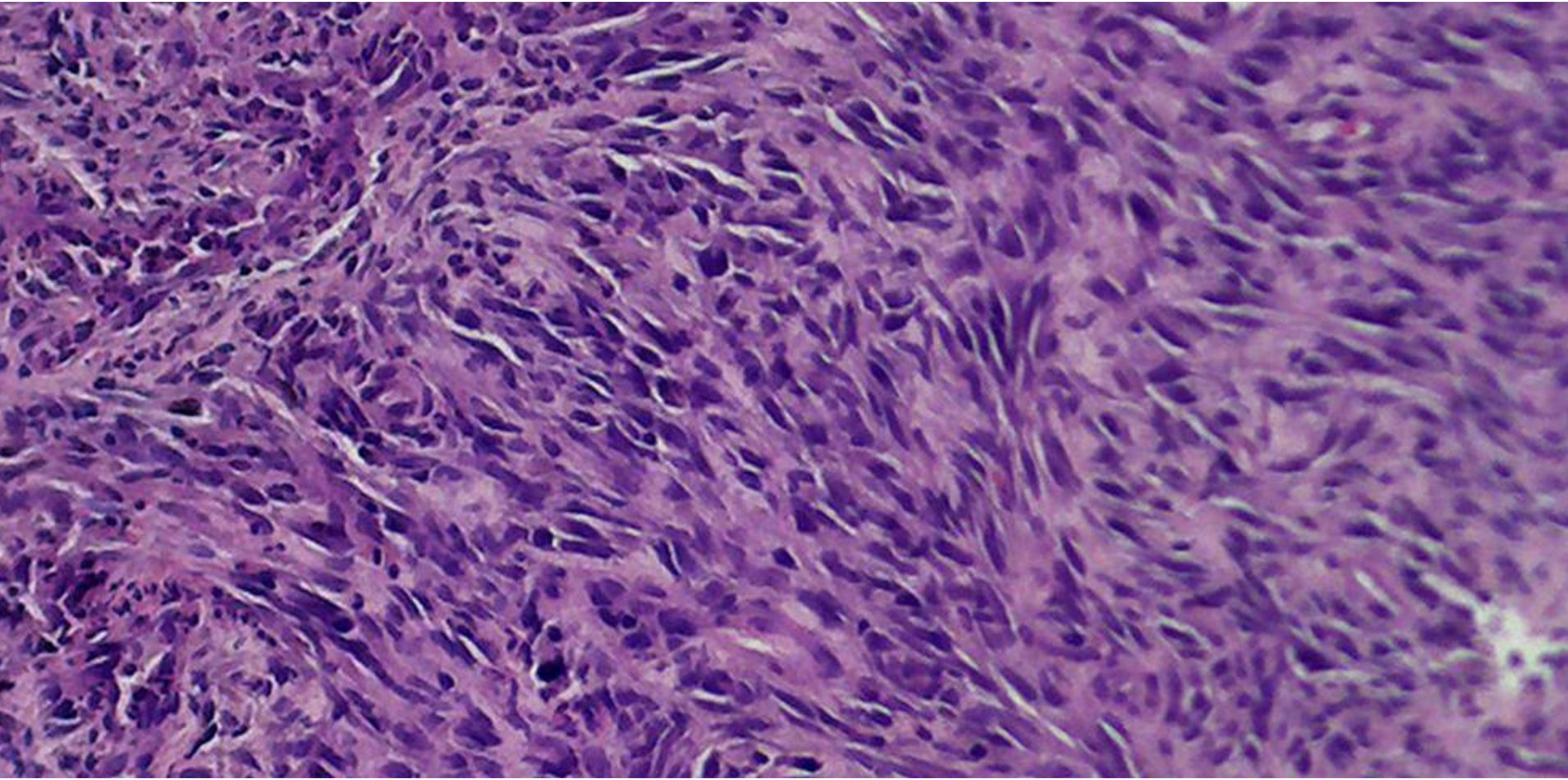
GASTRIC BIOPSY









DIFFERENTIALS

Gist

Leiomyoma

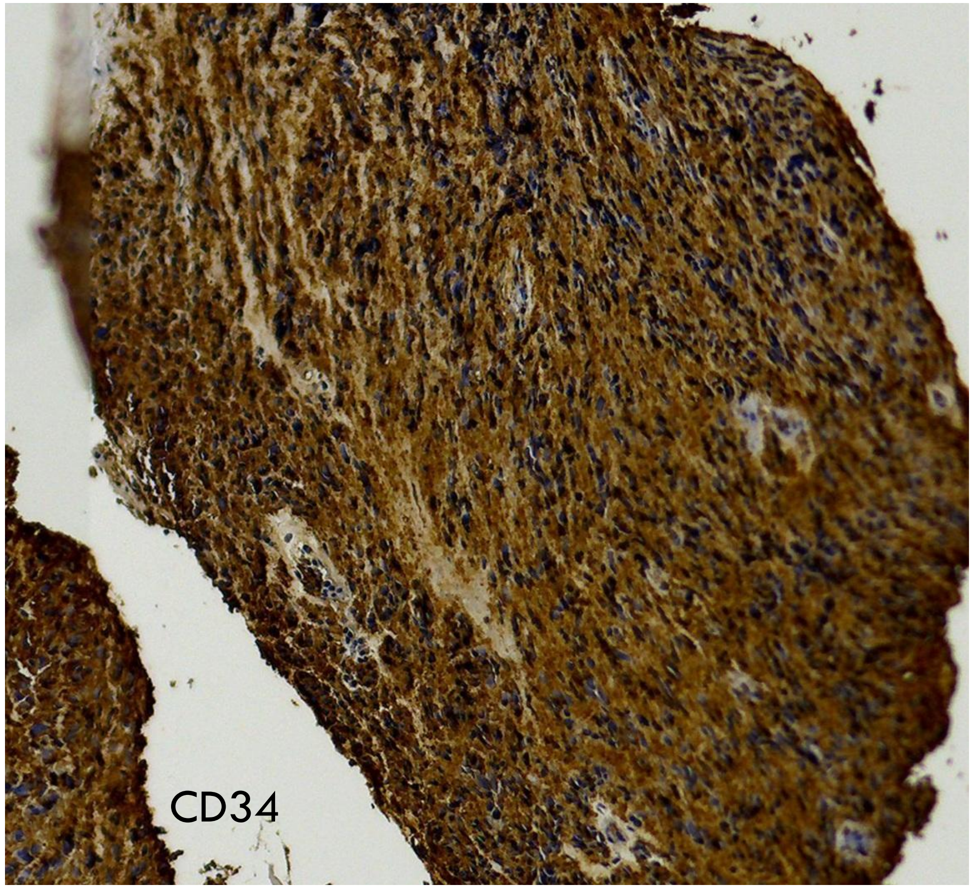
Shwannoma

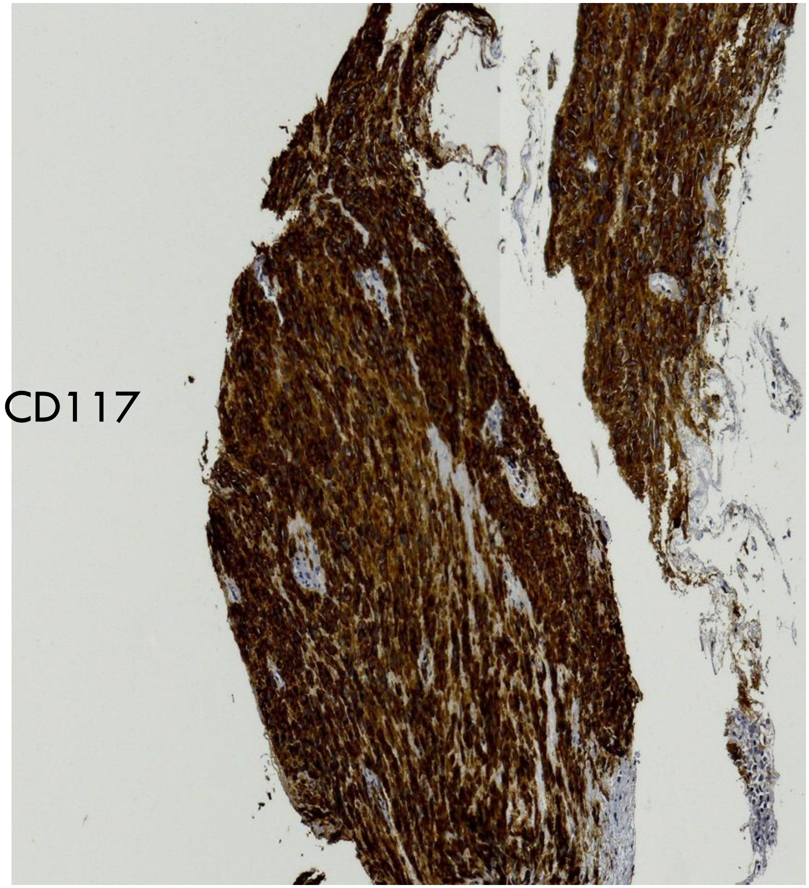
SFT

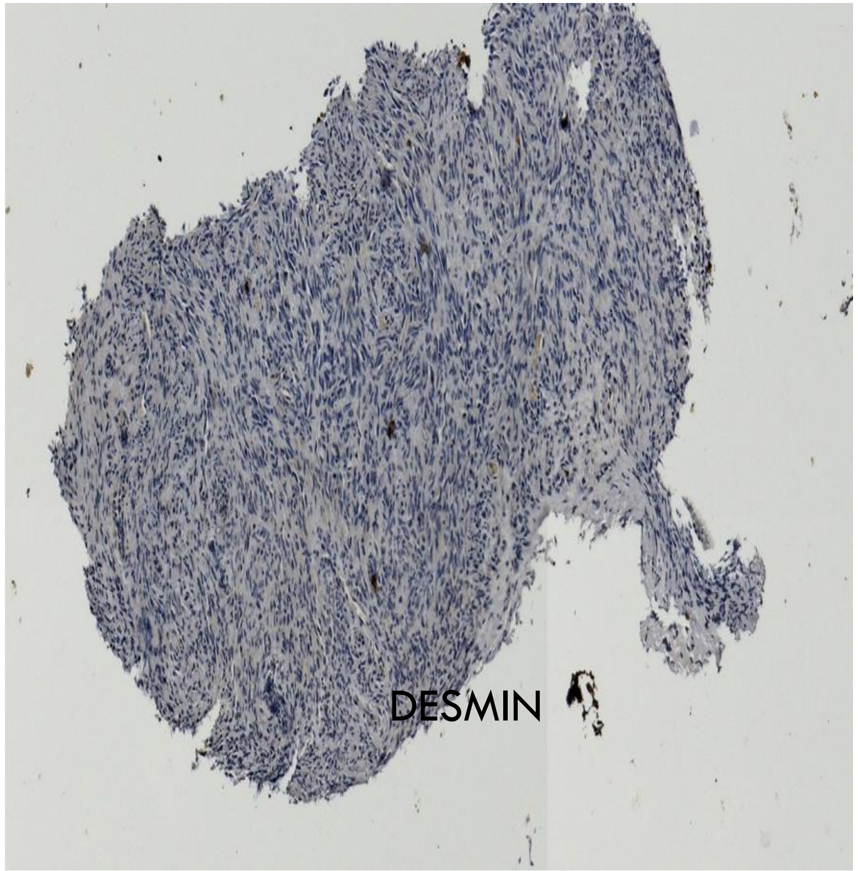
Inflammatory fibroid polyp

Desmoid fibromatosis

IMMUNOHISTOCHEMISTRY



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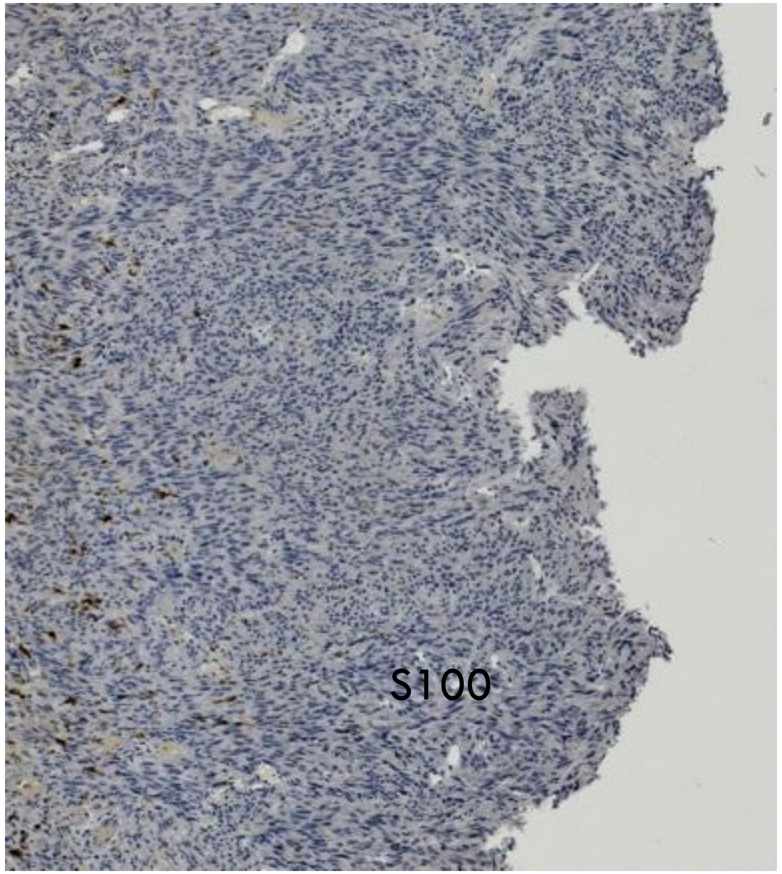

S100

DIAGNOSIS

Gist CD117 +ve, CD34 +ve, desmin –ve, S100 –ve.

Leiomyoma desmin +ve, CD117 –ve.

Shwannoma S100 +ve, CD117 -ve.

SFT STAT 6 +ve, CD34 +ve, BCL2 +ve, S100 –ve, CD117 –ve.

Inflammatory fibroid polyp CD34 +ve, variable ASMA, CD117 –ve, eosinophil rich.

GIST

GIST is the most common mesenchymal tumor of the gastrointestinal tract, arising from (or differentiating toward) the **interstitial cells of Cajal**—the pacemaker cells regulating gut peristalsis.

Activating mutations in:

- **KIT** (~75–80%)
- **PDGFRA** (~5–10%)

Wild-type GIST subsets include:

- SDH-deficient GIST
- BRAF-mutant or NF1-associated GIST

These mutations lead to **constitutive tyrosine kinase activation**

Anatomic Distribution; Stomach (~60%), Small intestine (~30%), Colon/rectum, esophagus (rare) Extra-GI sites (rare; "EGIST")

3 main morphologic patterns: **Spindle cell type 70%(most common), Epithelioid type 20%** (more common in PDGFRA-mutant GIST) & **Mixed type 10%**

GIST

Risk Stratification (Not benign vs malignant)

Tumor size, <2cm , >10cm

Mitotic rate (per 50 HPF)

Anatomic site

Key points:

Gastric GIST → better prognosis

Small intestinal GIST → more aggressive

High mitotic index → strongest predictor of adverse outcome

Always confirm with **CD117 + DOG1**

Not all GISTs are CD117 positive → DOG1 is critical

Prognosis = **size + mitoses + site**

Molecular testing is **standard of care**

Avoid labeling as “benign” → use **risk categories**

CASE 2

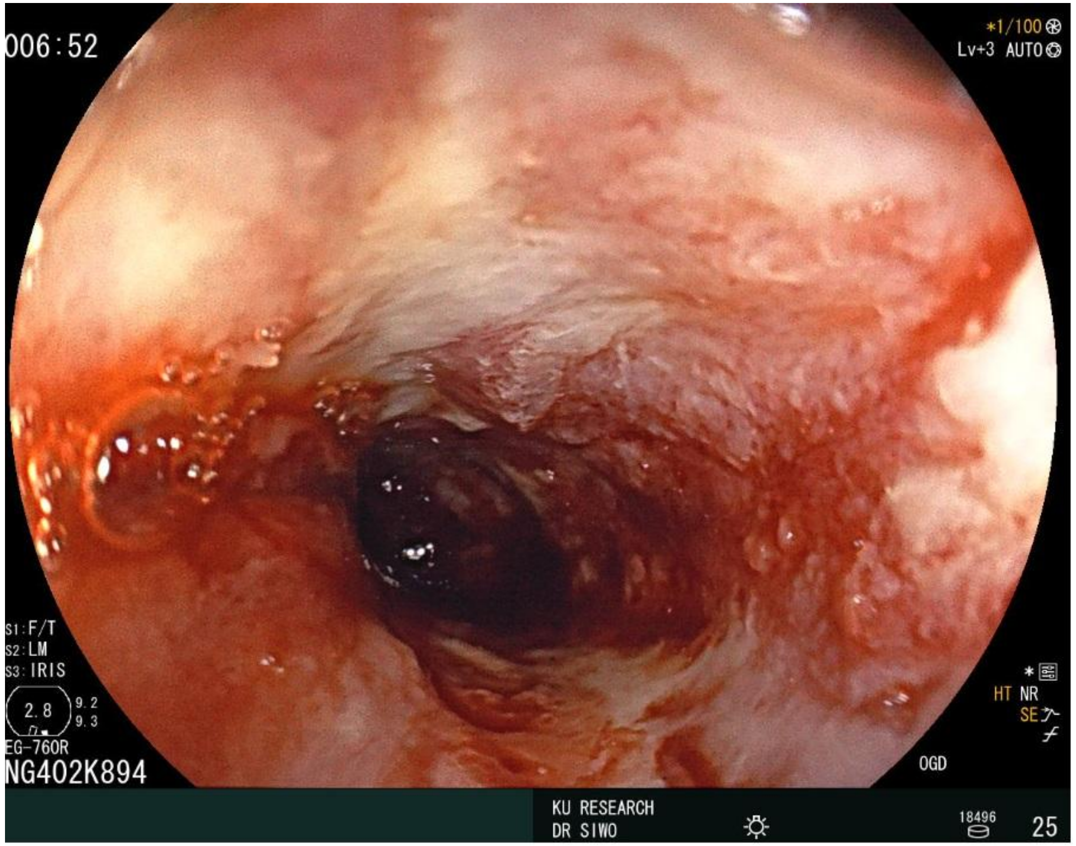
EW

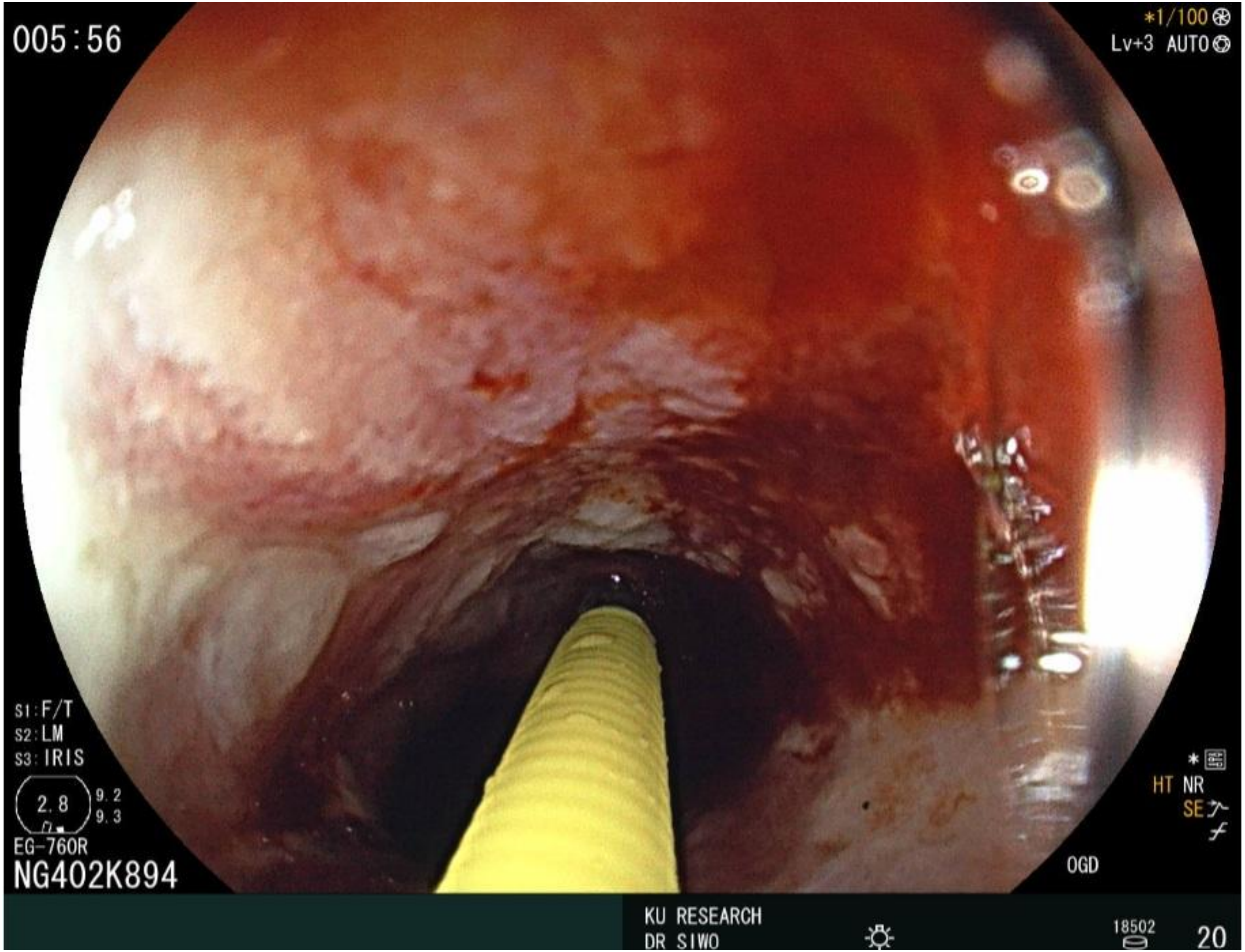
73yr Female

Presented with dysphagia and weightloss

Endoscopic findings of a sternotic mass

ENDOSCOPIC FINDING

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S3:IRIS
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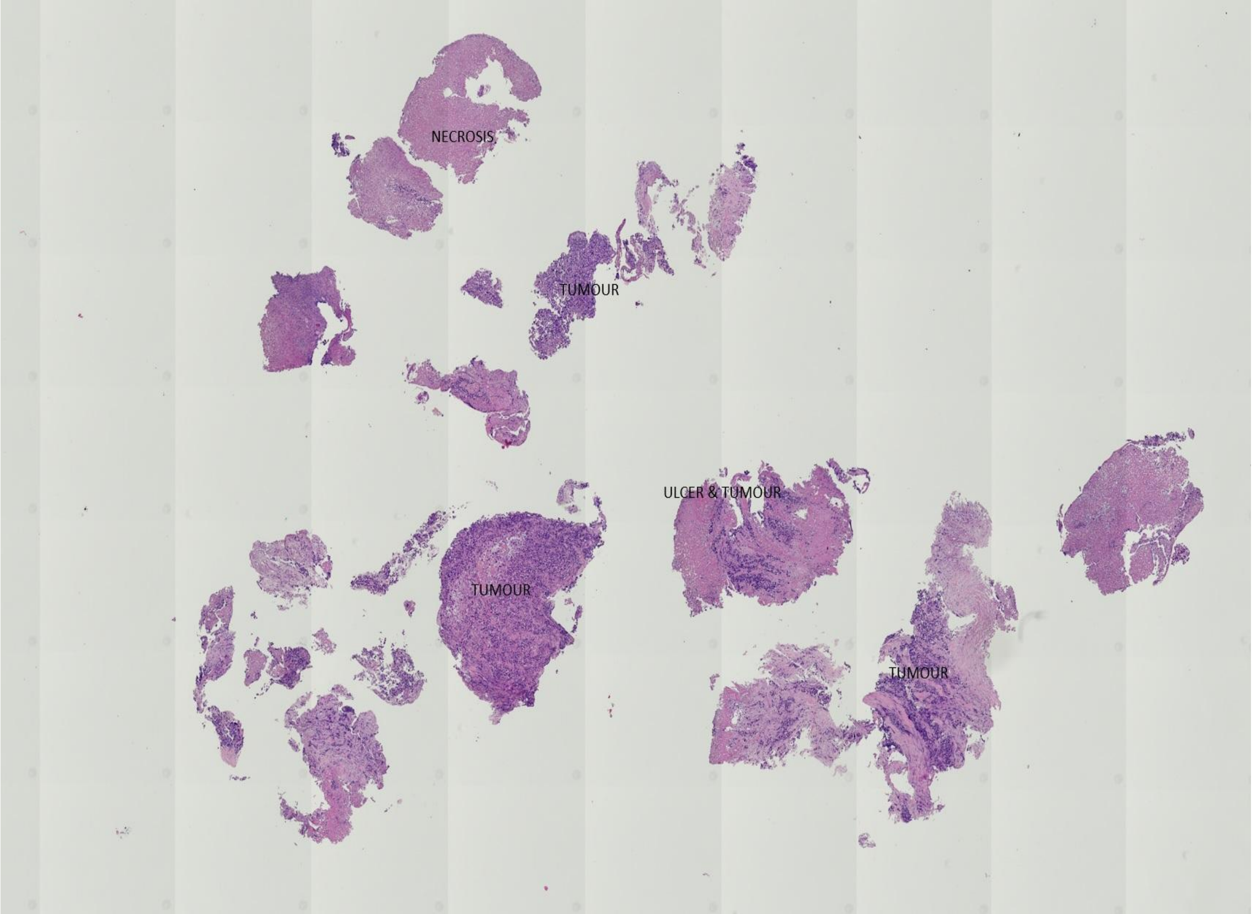
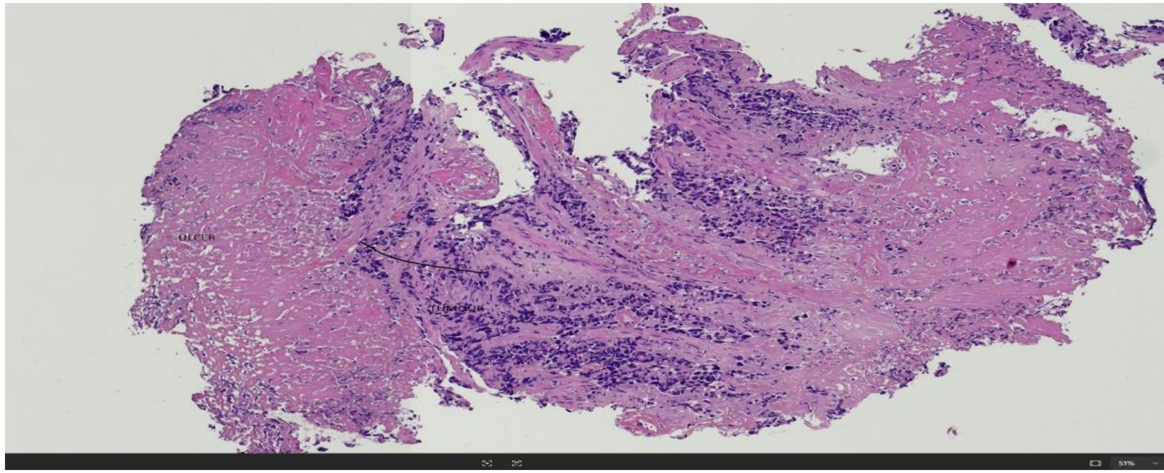
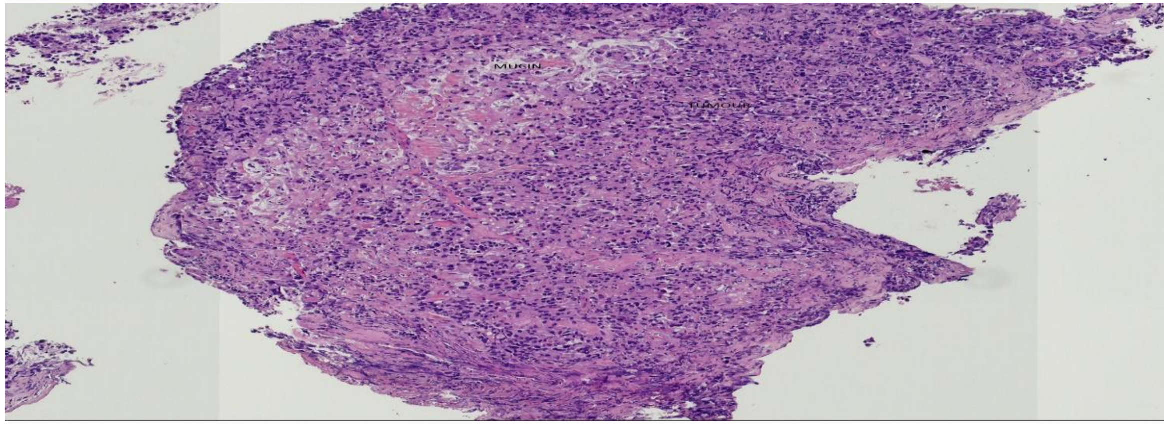
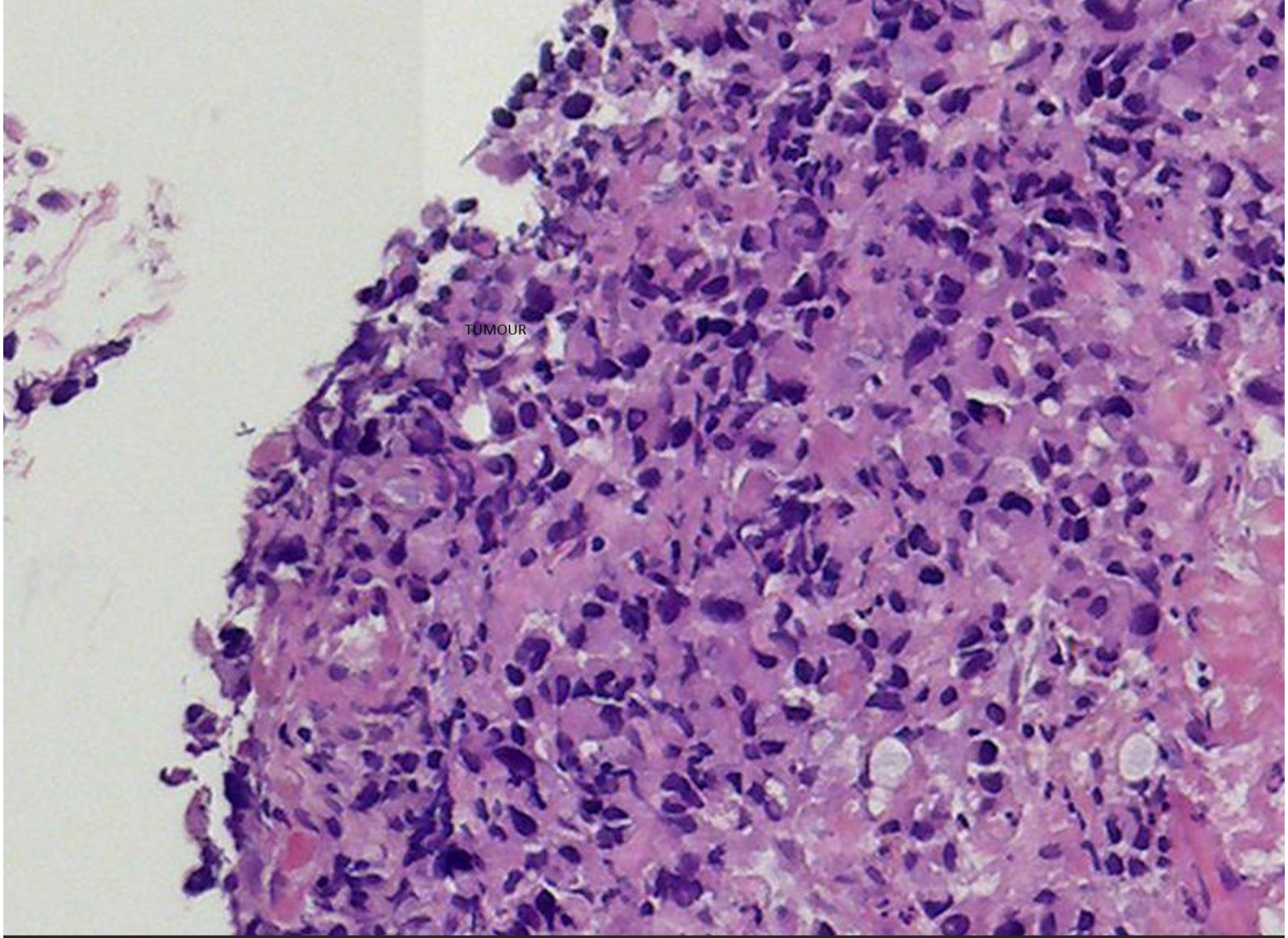
ESOPHAGEAL BIOPSY







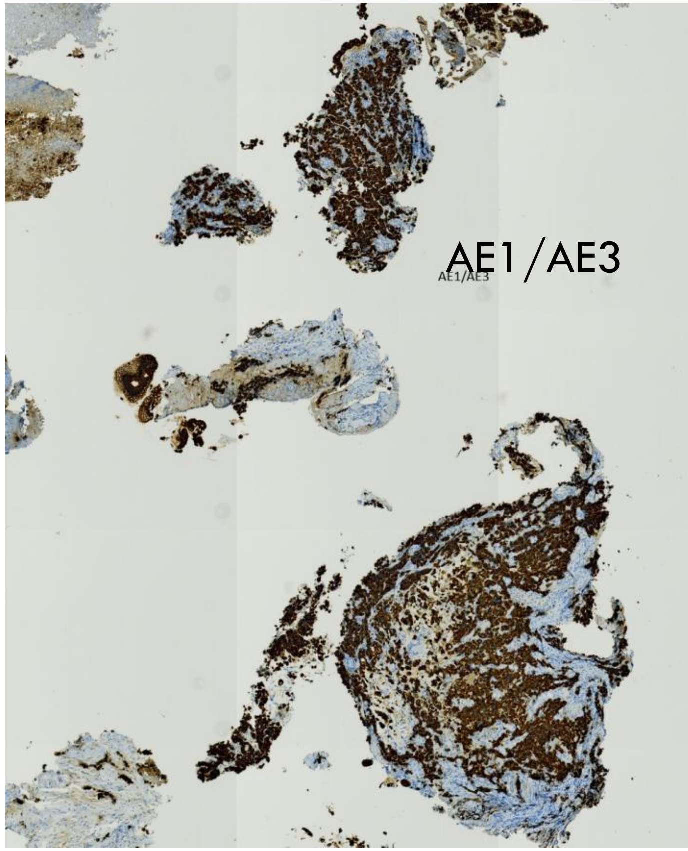

TUMOUR

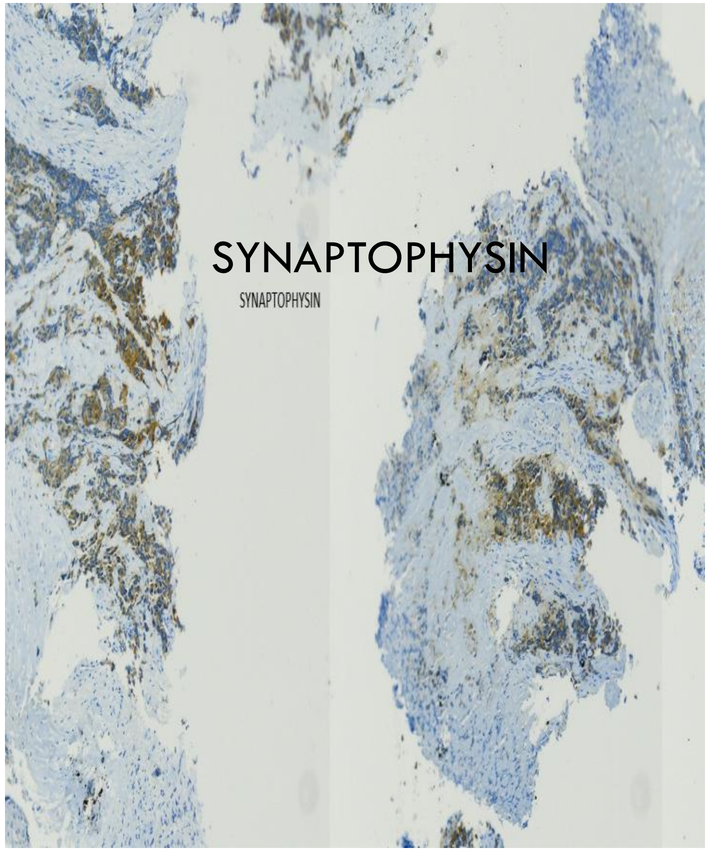
DIFFERENTIALS

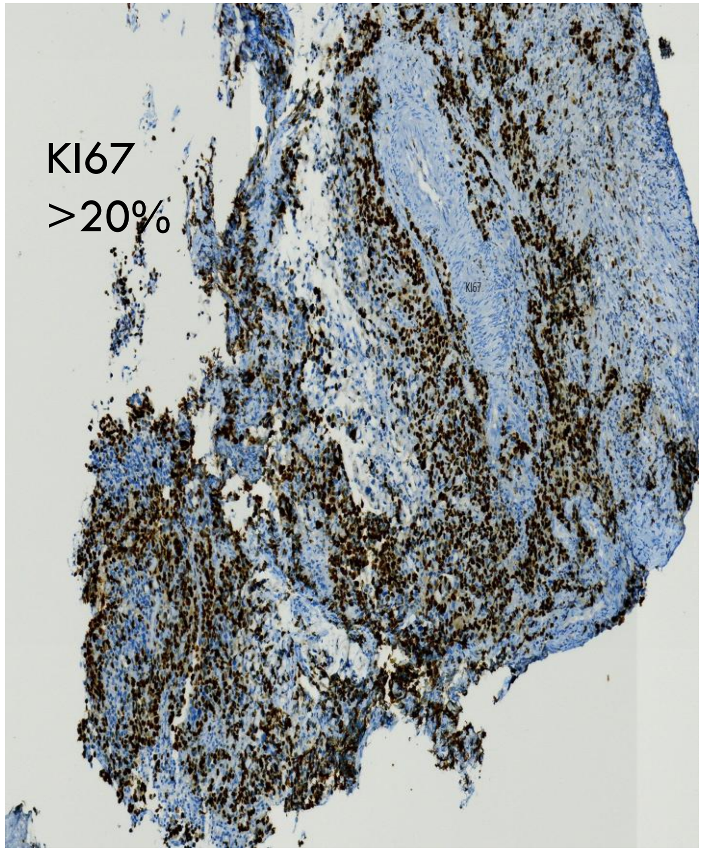
Poorly differentiated carcinoma

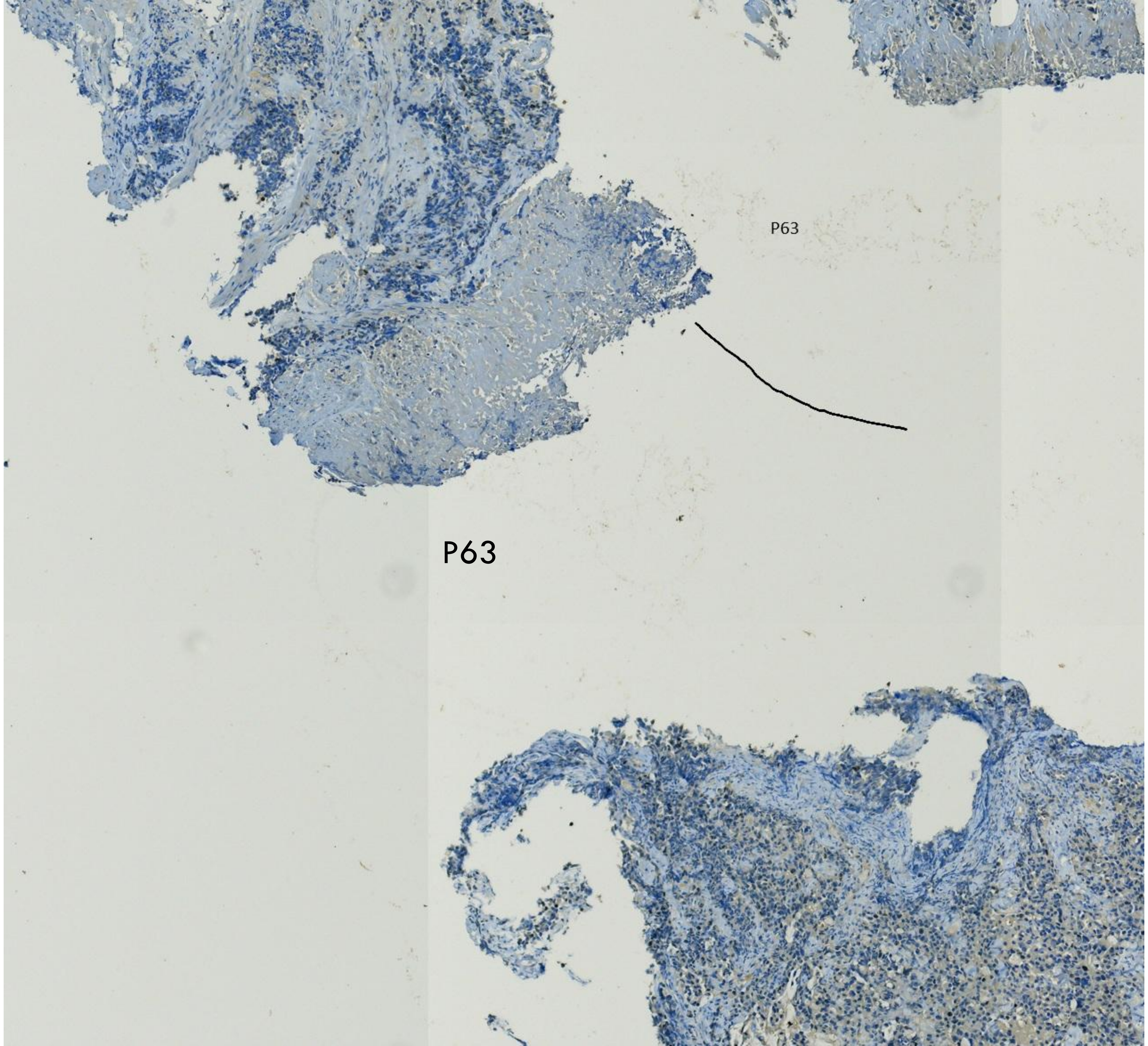
- Poorly differentiated neuroendocrine carcinoma
- Poorly differentiated squamous cell carcinoma

IMMUNOHISTOCHEMISTRY









P63

DIAGNOSIS

- High grade neuroendocrine carcinoma Synaptophysin +ve, AE1/AE3 +VE, Ki67 >20%, P63 -ve,

QUIZ

A 48-year-old man undergoes resection of a jejunal mass. Gross examination shows a 9 cm tumor with focal hemorrhage. Microscopy reveals a spindle cell neoplasm with a mitotic count of 12 per 50 high-power fields. Immunohistochemistry is positive for **CD117** and DOG1.
Which of the following factors **most strongly predicts aggressive clinical behavior** in this tumor?
A. Tumor location in the small intestine
B. Tumor size greater than 5 cm
C. High mitotic index (>5 per 50 HPF)
D. Spindle cell morphology
E. Expression of CD117

ANSWER

C. High mitotic index (>5 per 50 HPF)

Risk stratification in gastrointestinal stromal tumors (GISTs) is based on a combination of:

Tumor size

Mitotic activity

Tumor location

Among these, the **mitotic index is the single most powerful predictor of aggressive behavior,** as it directly reflects proliferative activity.

Q2

A 64-year-old man with progressive dysphagia and weight loss undergoes endoscopic biopsy of a distal esophageal mass. Histology reveals sheets of small, round to oval cells with scant cytoplasm, hyperchromatic nuclei, nuclear molding, and extensive necrosis. Numerous mitotic figures are identified (>40/10 HPF).

Immunohistochemistry shows strong positivity for **synaptophysin** and **chromogranin A**, with a Ki-67 proliferation index of approximately 85%. Tumor cells are also positive for cytokeratin.

Which of the following is the **most appropriate classification of this tumor according to WHO criteria?**

A. Well-differentiated neuroendocrine tumor (NET G1)
B. Well-differentiated neuroendocrine tumor (NET G2)
C. Well-differentiated neuroendocrine tumor (NET G3)
D. Poorly differentiated neuroendocrine carcinoma (small cell type)
E. Poorly differentiated neuroendocrine carcinoma (large cell type)

CORRECT ANSWER:

D. Poorly differentiated neuroendocrine carcinoma (small cell type)